

1. Does the Factory manage the lifecycle (close()) of the sensor?

No, the Factory does not close the lifecycle of the sensor. The Factory is only a creational pattern and the Factory's job ends when it returns the object to the client. The life cycle falls under operational tasks and should be managed by an operational class.

2. Should the Factory also implement retry logic?

No, the Factory should not implement any retry logic. The Factory cannot because it is for creation and not testing or managing. Implementing retry logic into the Factory would be making the creation class handle operational tasks.

3. What would happen if we added a third sensor in the future?

Adding another sensor would be difficult because the Factory relies on hardcoded keys rather than iterative ones. This means that adding a third sensor would break the Open/Closed Principle.